Kontrol Sistemleri

Accuracy of Systems: Steady-State Error (Ders 9)

Original notes: Control-Systems-lecture-9.pdf

UYARI Bu belge, el yazısı ders notlarından Claude (Anthropic) yardımıyla dijitale aktarılmıştır. Yardımcı kaynak olarak hazırlanmıştır; hata veya eksik içerebilir. Lütfen orijinal notları ve ders kaynaklarını esas alın.

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orig. p. 1

Ders 9

Accuracy of Systems

The ability of the system follow the given instruction can be used as a measure of classification.

* Common input: step, ramp, parabolic

$$u(t) = 1 \quad , \quad r(t) = t \quad , \quad P(t) = t^2$$

Figure: Three small sketches under the equations, each with axes: step (constant level), ramp (straight line rising from the origin) and parabola (convex curve rising from the origin). Orange note under the step sketch: "tracking systems". At the right, a sketch of a ground antenna (red circle) with orbit arcs: an orange low arc labelled "LEO", dashed higher arcs labelled "35 750 km" and "Geo", and "sat" at the satellite.
(in the original notes, p. 1)

* Comparison of accuracy is possible by using the steady-state error magnitude!!

if $\zeta < 1$ in a second o.s

Figure: Two stacked sketches against t. Upper: c(t), an underdamped step response with overshoot that oscillates and settles to a constant value. Lower: e(t), the error, starting at a peak and oscillating about zero while decaying.
(in the original notes, p. 1)



Figure: Feedback block diagram: R(s) enters a summing junction (plus on the input, minus on the feedback); the junction output goes through block G(s) to the output C(s); a feedback path from the output through block H(s) returns to the junction. An arrow from the junction output carries the annotation given next.
(in the original notes, p. 1)

$$E(s) = R(s) - C(s)H(s)$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

$$E(s) = R(s) - G(s)H(s) \Rightarrow \frac{E(s)}{R(s)} = 1 - \frac{G(s)H(s)}{R(s)}$$

$$\Rightarrow \frac{E(s)}{R(s)} = 1 - \frac{G(s)H(s)}{1 + G(s)H(s)} = \frac{1}{1 + G(s)H(s)}$$

$$e(\infty) = e_{ss} = \lim_{t\to\infty} e(t) \overset{\text{Final value theorem}}{=} \lim_{s\to 0} sE(s)$$

$$= \lim_{s\to 0} \frac{sR(s)}{1 + G(s)H(s)}$$

orig. p. 2

Validity of final value theorem

If the poles are in zone 1 so it can't have final value

Only works with Zone III and zone IV

Stability → CLP ✓

OLP



Figure: s-plane sketch: horizontal axis Re (arrow), vertical axis Im drawn as a thick line. Right half-plane bounded by red dashes and labelled "Zone 1" (red), with "unstable" and an arrow at the bottom right; two crosses (poles) in the right half-plane close to the Im axis, one above and one below the Re axis. Left half-plane bounded by blue dashes and labelled "Zone III" (blue), with "stable" (blue) at the bottom left. The Im axis is coloured orange, labelled "Zone II" with an arrow to "marginally stable" below it. A purple arrow from the origin towards the lower left is labelled "Zone IV".
(in the original notes, p. 2)

Zone I :

$$G(s) = \text{T.F} = \frac{1}{s - 2}$$

$$\text{FVT} = \lim_{s\to 0} sG(s) = \lim_{s\to 0} \frac{s}{s - 2} = 0 \quad \times \text{ wrong}$$

but

> *Figure:* Sketch of c(t) against t: a curve starting at the origin and growing upward exponentially (convex).
> *(in the original notes, p. 2)*

For Zone II

$$\text{T.F} = G(s) = \frac{1}{s^2 + 2}$$

$$s_{1,2} = \pm j\sqrt{2}$$

$$\text{FVT}: \lim_{s\to 0} sG(s) = \lim_{s\to 0} \frac{s}{s^2 + 2} = 0 \quad \times$$

> *Figure:* Sketch of a sustained sinusoidal oscillation of constant amplitude about the time axis, with dashed horizontal lines marking the upper and lower amplitude levels.
> *(in the original notes, p. 2)*

Zone III

$$\text{T.F} = G(s) = \frac{1}{s + 2} \quad \Rightarrow s = -2$$

$$\text{FVT}: \lim_{s\to 0} \frac{s}{s + 2} = 0$$

> *Figure:* Sketch labelled e^{-2t}: a decaying exponential starting high at t=0 and approaching zero along the time axis.
> *(in the original notes, p. 2)*

orig. p. 3

Zone IV :

> *Figure:* s-plane sketch (Im vertical axis, Re horizontal axis) with a cross (pole) at the origin; label "pole Location".
> *(in the original notes, p. 3)*

$$\text{TF} = \frac{1}{s}$$

$$\text{FVT} = \lim_{s\to 0} s\frac{1}{s} = 1$$

$$\lambda = 0$$

$$e^{\lambda t} = e^{0t} = 1$$

Figure: Sketch with vertical and horizontal axes: a constant level (horizontal line above the time axis) starting at the vertical axis.
(in the original notes, p. 3)

Figure: Block diagram: R(s) (plus) enters a summing junction (minus on the feedback), E(s) leaves the junction into block G(s), output C(s); unity feedback path from the output straight back to the junction.
(in the original notes, p. 3)

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$$

$G(s)$: open loop TF for unity feedback

$$\frac{E(s)}{R(s)} = \frac{1}{1 + G(s)}$$

Figure: Two block diagrams joined by the sign ≡. Left: R (plus) into a summing junction (minus on the feedback), E after the junction, block G(s), output C, feedback block H(s) from the output back to the junction. Right: R(s) (plus) into a summing junction, one block labelled "Ge", output C(s), unity feedback path.
(in the original notes, p. 3)

$$G(s) = \frac{k\,(T_a s + 1)(T_b s + 1)\cdots(T_m s + 1)}{s^N (T_1 s + 1)(T_2 s + 1)\cdots(T_p s + 1)}$$

$P_{1,\ldots,N} = 0 \leftarrow$ N multiplicity $\rightarrow$ integrator

$p = \frac{-1}{T_1}, \frac{-1}{T_2}, \cdots$

if N=0 $\rightarrow$ type zero

if N=1 $\rightarrow$ Type one

⋮

In most cases the most integrator is 3

orig. p. 4

Accuracy of systems

case 1) step input: $R(s) = \frac{1}{s}$

$$\frac{E(s)}{R(s)} = \frac{1}{1+G(s)}$$

$$e_{ss} = \lim_{s\to 0} \frac{sR(s)}{1+G(s)}$$

$$e_{ss} = \lim_{s\to 0} \frac{1}{1+G(s)} = \frac{1}{1+\lim_{s\to 0} G(s)} = \frac{1}{1+k_p}$$

? constant

$$k_p = \lim_{s\to 0} G(s) \qquad \text{(static position error)} \qquad \text{DC gain}$$

$$\text{Type 0:} \quad G(s) = \frac{k(T_a s+1)(T_b s+1)\cdots(T_m s+1)}{s^N (T_1 s+1)(T_2 s+1)\cdots(T_p s+1)}$$

$$N = 0 \Rightarrow k_p = \lim_{s\to 0} G(s) = k$$

$$N = 1 \Rightarrow k_p = \lim_{s\to 0} G(s) = \infty$$

$$N = 2 \Rightarrow k_p = \lim_{s\to 0} = \infty$$

$$N \geq 1 \Rightarrow k_p = \infty$$

case 2) input is ramp

$$R(s) = \frac{1}{s^2}$$

$$e_{ss} = \lim_{s\to 0} \frac{s\frac{1}{s^2}}{1+G(s)} = \lim_{s\to 0} \frac{1}{s(1+G(s))}$$

$$= \frac{1}{\lim_{s\to 0} sG(s)} = \frac{1}{k_v}$$

k_v = static velocity error = $\lim_{s\to 0} sG(s)$

$$\text{Type 0:} \quad k_v = \lim_{s\to 0} sG(s)$$

$$= \lim_{s\to 0} \frac{s\,k(T_a s+1)(T_b s+1)\cdots(T_m s+1)}{(T_1 s+1)(T_2 s+1)\cdots(T_p s+1)} = 0$$

$$\text{Type 1:} \quad k_v = \lim_{s\to 0} sG(s) = \lim_{s\to 0} \frac{s\,k(T_a s+1)\cdots}{s(T_1 s+1)} = k$$

$$\text{Type 2:} \quad k_v = \lim_{s\to 0} \frac{s\,k(T_a s+1)\cdots}{s^2(T_1 s+1)\cdots} = \infty$$

case 3)

$$R(s) = \frac{1}{s^3}$$

$$e_{ss} = \lim_{s\to 0} \frac{s\frac{1}{s^3}}{1+G(s)} = \frac{1}{\lim_{s\to 0} s^2 G(s)} = \frac{1}{k_a} \rightarrow \text{static acceleration error}$$

$$N = 0: \quad k_a = \lim_{s\to 0} \frac{s^2 k(T_a s+1)(T_b s+1)\cdots}{(T_1 s+1)\cdots} = 0$$

orig. p. 5

	input		
Type	step	ramp	parabolic
0	$\frac{1}{1+k}$	∞	∞
1	0	$\frac{1}{k}$	∞
2	0	0	$\frac{1}{k}$
3	0	0	0

Example :

Figure: Block diagram: R(s) (plus) into a summing junction (minus on the feedback), then a block k, then a block 1/(s(s+1)(s+2)), output C(s); unity feedback path from the output back to the junction.
(in the original notes, p. 5)

$$\text{OL T.F} = \frac{k}{s(s+1)(s+2)} \qquad \text{Type 1 system}$$

Figure: s-plane sketch (Im vertical axis, Re horizontal axis) with a cross at the origin.
(in the original notes, p. 5)

$$\frac{E(s)}{R(s)} = \frac{1}{1+G(s)}$$

$$e_{ss} = \lim_{s\to 0} \frac{sR(s)}{1+G(s)}$$

$$e_{ss} = \lim_{s\to 0} \frac{s\cdot\frac{1}{s^2}}{1+\frac{k}{s(s+1)(s+2)}} = \lim_{s\to 0} \frac{s(s+1)(s+2)}{s\left[s(s+1)(s+2)+k\right]}$$

$$= \frac{2}{k} = 0{,}01 \Rightarrow k = 200 \checkmark$$

$\uparrow$ for $e_{ss} = 0{,}01$ design

But we have to check stability

$$\frac{\frac{k}{s(s+1)(s+2)}}{1+\frac{k}{s(s+1)(s+2)}} = \frac{k}{s(s+1)(s+2)+k}$$

orig. p. 6

Stability

$k \to p$

$$s(s+1)(s+2)+k=0$$
$$s(s^2+3s+2)+k=0$$
$$s^3+3s^2+2s+k=0$$

s^3	1	2	
s^2	3	k	$\Rightarrow 6-k \geq 0 \Rightarrow k \leq 6$
s^1	$\frac{6-k}{3}$	0	$k \geq 0$
s^0	k		$0 \leq k \leq 6$

$k = 6$ is the max possible k here !!!

so max $e_{ss} = 0{,}33$ because the system have to be stable because final value theorem is not working with unstable systems.

Steady state error for disturbance

Figure: Block diagram: the input (plus) enters a first summing junction (minus on the feedback from the output); block $G_1(s)$ labelled "controller block"; its output (plus) enters a second summing junction where the disturbance $D(s)$ (plus) enters from above; then block $G_2(s)$ labelled "plant"; output $C(s)$; unity feedback path from the output back to the first junction.
(in the original notes, p. 6)

Figure: Sketch labelled "plane": side view of a tail/wing with a movable control surface labelled "ruben" (ruder) and a smaller "trim tab" on it.
(in the original notes, p. 6)

$$C(s) = \left[E(s)G_1(s) + D(s)\right] G_2(s)$$
$$E(s) = R(s) - C(s) \Rightarrow C(s) = R(s) - E(s)$$
$$E(s) = \frac{R(s)}{1+G_1(s)G_2(s)} - \frac{G_2(s)D(s)}{1+G_1(s)G_2(s)}$$

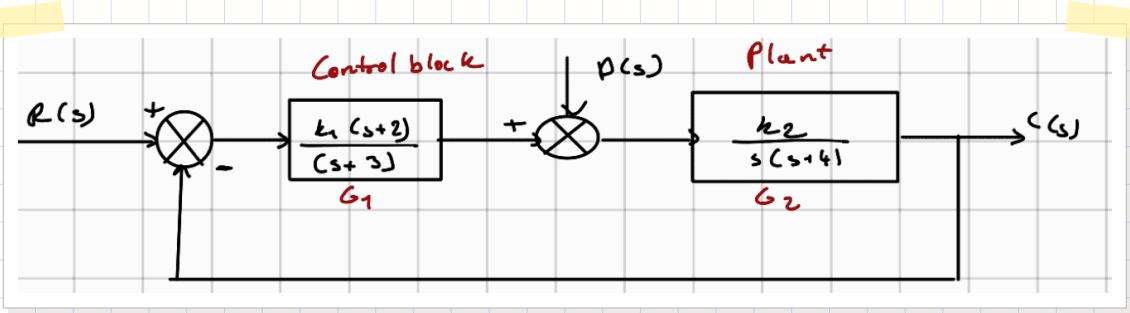
$$e_{ss} = \lim_{s\to 0} sE(s)$$
$$= \underbrace{\lim_{s\to 0} \frac{sR(s)}{1+G_1G_2}}_{e_R(\infty)} - \underbrace{\lim_{s\to 0} \frac{sG_2D}{1+G_1G_2}}_{e_D(\infty)}$$

orig. p. 7

because it is negative we can decrease error with it

Example

Design k_1 and k_2 so that
1 - e_{ss} due to unit step disturbance : $-0{,}00001$
2 - e_{ss} " " ramp input : $0{,}002$



original drawing, p. 7

$$\frac{k_1(s+3-1)}{s+3} = \frac{k_1(s+3)}{s+3} - \frac{k_1}{s+3} = k_1 - \frac{k_1}{s+3}$$

$$e_{ss}^{D} = -\lim_{s\to 0} \frac{s\,G_2(s)D(s)}{1+G_1(s)G_2(s)} = -\lim_{s\to 0} \frac{G_2(s)}{1+G_1(s)G_2(s)} = \frac{-3}{2k_1}$$

$$-0{,}00001 = \frac{-3}{2k_1} \Rightarrow k_1 = 150\,000$$

$$e_{ss}^{R} = \lim_{s\to 0} \frac{sR(s)}{1+G_1(s)G_2(s)} = \lim_{s\to 0} \frac{1}{s\,(1+G_1(s)G_2(s))}$$

$$= \frac{6}{k_1k_2} = 0{,}002 \Rightarrow k_2 = 0{,}02$$

$$\text{Total } e_{ss} = \frac{6}{k_1k_2} - \frac{3}{2k_1}$$

$$= 0{,}02^{1} - 0{,}00001$$

this system order is 3 by finding characteristic equation.

Sensitivity

The degree to which changes in the system parameters affect system T.F. and hence performance (error, stability, ...) of the system

orig. p. 8

Example

$$F = \frac{k}{k+a} \qquad \text{if } k = 10$$

[1] Claude notu: muhtemelen 0,002 – 0,00001 olmalı.

$$F = \frac{10}{10+a} \qquad \begin{cases} a = 100 \rightarrow F = \dfrac{10}{110} = 0{,}09 \\ \quad (+200\%) \\ a = 300 \rightarrow F = \dfrac{10}{310} = 0{,}032 \end{cases} \qquad \curvearrowright -65\%$$

$$\frac{\Delta F}{F} \times 100 = \frac{0{,}032 - 0{,}091}{0{,}091} \times 100 = -65\%$$

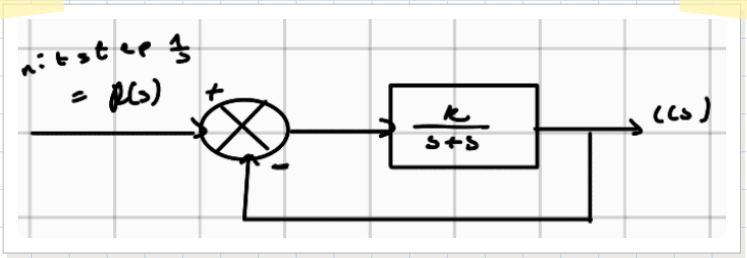
$$\frac{\Delta a}{a} \times 100 = \frac{300 - 100}{100} \times 100 = 200\%$$

(Reduced sensitivity)
This means it's low sensitive to a
Average sensitivity : $\frac{65\%}{200\%} < 1$ (Reduced sensitivity)

$$S_{F:a} = \lim_{\Delta p \to 0} \frac{\text{fractional change in function}(F)}{\text{" " " parameter}(p)}$$

$$= \lim_{\Delta p \to 0} \frac{\Delta F/F}{\Delta p/p} = \lim_{\Delta p \to 0} \frac{p}{F}\frac{\Delta F}{\Delta p}$$

$$S_{F:a} = \frac{p}{F}\frac{\partial F}{\partial p}$$



original drawing, p. 8

Check the sensitivity of ss. error w.r.t. parameter 'k'

$$e(\infty) = e_{ss} = \lim_{s \to 0} \frac{sR(s)}{1 + G(s)} = \lim_{s \to 0} \frac{1}{1 + G(s)}$$

$$= \frac{1}{1 + \lim\limits_{s \to 0} \frac{k}{s+5}} = \frac{5}{5+k}$$

$$S_{e:k} = \frac{k}{e_{ss}} \cdot \frac{\partial e_{ss}}{\partial k} = \left(\frac{k}{\frac{5}{5+k}}\right)\left(-\frac{5}{(5+k)^2}\right)$$

$$= \frac{k(5+k)}{5}\left(-\frac{5}{(5+k)^2}\right) = -\frac{k}{5+k}$$

for $k = 10 \Rightarrow S_{e:k} = \frac{-10}{5+10} = \frac{-10}{15} = \frac{-2}{3}$

if 'k' is changed to 12

$$\frac{12 - 10}{10} \leftarrow \frac{\Delta k}{k} \times 100 = 20\%$$

$\delta \rightarrow$ delta also

$$\underbrace{S_{e:k}}_{-\frac{2}{3}} \cdot \underbrace{\frac{\Delta k}{k}}_{\frac{1}{5}} = \frac{\delta e_{ss}}{e_{ss}} = -\frac{2}{3} \times \frac{1}{5} = \frac{-2}{15} = -13\%$$

orig. p. 9

$$\delta e_{ss} = -0{,}13 \times e_{ss}$$

$$= -0{,}13 \times \frac{1}{3} = -0{,}04 \qquad \text{the amount of change}$$